

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q15-0428

Auburn Watt Tap 1 138 kV  
July 22, 2016

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0428. The Interconnection Customer proposes a 203.6 MW solar photovoltaic facility interconnecting at the Auburn Watt Tap 1 138 kV substation in the Pennsylvania area. The request was received on January 24, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 225 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$82,621,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                               | Contingency                                                               | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------------------------------|---------------------------------------------------------------------------|-------------|--------------|----------|
| OSWEGO HARBOR POWER-LARENCE LIVERMORE 138 KV                     | CURECANTI-MOSER GENERATING STA 345 KV                                     | 169.4       | 305          | 31.6     |
| LAWRENCE LIVERMORE-ELROY 138 KV                                  | CURECANTI-MOSER GENERATING STATION 345 KV                                 | 164.1       | 294          | 31.6     |
| AUBURN WATT TAP 1-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV        | NUCOR-WILSON DAM 138 KV                                                   | 160.3       | 260          | 34.9     |
| OSWEGO HARBOR POWER-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV line | CURECANTI-MOSER GENERATING STATION 345 KV                                 | 153.8       | 340          | 31.6     |
| ROTHADEW-ELROY 138 KV Line                                       | CHEROKEE FALLS-BANDERA ROD SUBSTATION 138 KV                              | 131.6       | 268          | 11.4     |
| PORTSIDE ENERGY-WAPATO 138 KV                                    | THETFORD-UNION CARBIDE 138 KV                                             | 129.3       | 162          | 6.6      |
| ROTHADEW SUBSTATION-BOWLING GREEN PEAKING 138 KV                 | CHEROKEE FALLS-BANDERA RD 138 KV                                          | 129.3       | 260          | 11.4     |
| WENTZVILLE ASSEMBLY & CONTIGUOUS-PARR RD 138 KV                  | AUBURN WATT TAP 1-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV Line            | 129.3       | 205          | 10.5     |
| PARR ROAD SUBSTATION-GRONMEYER 138 KV L                          | WENTZVILLE ASSEMBLY & CONTIGUOUS-AUBURN WATT TAP 1 SUBSTATION 138 KV Line | 126.4       | 221          | 10.5     |
| PARKER-GRONMEYER 138 KV Line                                     | WENTZVILLE ASSEMBLY & CONTIGUOUS-AUBURN WATT TAP 1 138 KV                 | 125.7       | 238          | 10.5     |

| Monitored Facility                       | Contingency                                                          | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------|----------------------------------------------------------------------|-------------|--------------|----------|
| PARER-NUCOR 138 KV Line                  | AUBURN WATT TAP 1-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV Line       | 124.6       | 247          | 10.5     |
| ELIZA CREEK-ONTELAUNEE ENERGY CTR 345 KV | TOMBSTONE-BIXBY HILL SUBSTATION 500 KV line                          | 124.2       | 1,617        | 5.5      |
| NUCOR-WILSON DAM 138 KV                  | AUBURN WATT TAP 1-WENTZVILLE ASSEMBLY & CONTIGUOUS SUBSTATION 138 KV | 123.9       | 262          | 10.5     |
| THETFORD-UNION CARBIDE 138 KV line       | ALBANY-FAIRGROUNDS 138 KV L                                          | 122.8       | 209          | 7.7      |
| ALBANY-FAIRGROUNDS 138 KV                | THETFORD-UNION CARBIDE SUBSTATION 138 KV                             | 122.5       | 243          | 6.6      |
| FAIRGROUNDS-IONA 138 KV                  | UNION CARBIDE-THETFOD 138 KV                                         | 121.9       | 223          | 6.6      |
| BOWLING GREEN PEAKING-THETFORD 138 KV    | CHEROKEE FALLS-BANDERA RD 138 KV Line                                | 118.0       | 275          | 11.4     |
| WAPATO-IONA 138 KV line                  | THETFORD-UNION CARBIDE 138 KV                                        | 116.5       | 191          | 6.6      |
| UNION CARBIDE-JARRETT 138 KV             | ALBANY-FAIRGROUNDS 138 KV                                            | 112.5       | 204          | 7.7      |
| ELIZA CREEK-CURECANTI 345 KV             | OSWEGO HARBOR POWER-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV L        | 102.5       | 1,640        | 16.1     |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| OSWEGO HARBOR POWER-LAWRENCE LIVERMORE 138 KV                    | Rebuild line with higher-capacity conductor | \$8,850,000   | 64% (\$5,688,000)     |
| LAWRENCE LIVERMORE-ELROY 138 KV                                  | Rebuild line with higher-capacity conductor | \$5,747,000   | 64% (\$3,693,000)     |
| AUBURN WATT TAP 1-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV        | Rebuild line with higher-capacity conductor | \$6,524,000   | 100% (\$6,524,000)    |
| OSWEGO HARBOR POWER-WENTZVILLE ASSEMBLY & CONTIGUOUS 138 KV line | Rebuild line with higher-capacity conductor | \$18,896,000  | 64% (\$12,145,000)    |
| ELROY-ROTHADEW 138 KV                                            | Rebuild line with higher-capacity conductor | \$2,274,000   | 100% (\$2,274,000)    |
| WAPATO-PORTSIDE ENERGY 138 KV                                    | Rebuild line with higher-capacity conductor | \$2,575,000   | 100% (\$2,575,000)    |
| BOWLING GREEN PEAKING-ROTHADEW 138 KV                            | Rebuild line with higher-capacity conductor | \$7,423,000   | 100% (\$7,423,000)    |
| WENTZVILLE ASSEMBLY & CONTIGUOUS-PARR RD 138 KV                  | Rebuild line with higher-capacity conductor | \$2,167,000   | 100% (\$2,167,000)    |

| Facility                                  | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------|---------------------------------------------|---------------|-----------------------|
| GRONMEYER-PARR ROAD SUBSTATION 138 KV     | Rebuild line with higher-capacity conductor | \$11,453,000  | 100% (\$11,453,000)   |
| GRONMEYER-PARKER 138 KV                   | Rebuild line with higher-capacity conductor | \$4,500,000   | 100% (\$4,500,000)    |
| NUCOR-PARKER 138 KV                       | Rebuild line with higher-capacity conductor | \$8,580,000   | 100% (\$8,580,000)    |
| ELIZA CRK-ONTELAUNEE ENERGY CTR 345 KV    | Rebuild line with higher-capacity conductor | \$14,483,000  | 2% (\$307,000)        |
| NUCOR-WILSON DAM 138 KV                   | Rebuild line with higher-capacity conductor | \$2,718,000   | 100% (\$2,718,000)    |
| THETFORD-UNION CARBIDE 138 KV line        | Rebuild line with higher-capacity conductor | \$2,576,000   | 100% (\$2,576,000)    |
| ALBANY-FAIRGROUNDS 138 KV                 | Rebuild line with higher-capacity conductor | \$3,275,000   | 100% (\$3,275,000)    |
| FAIRGROUNDS-IONA 138 KV                   | Rebuild line with higher-capacity conductor | \$841,000     | 100% (\$841,000)      |
| BOWLING GREEN PEAKING-THETFORD 138 KV     | Rebuild line with higher-capacity conductor | \$1,059,000   | 100% (\$1,059,000)    |
| WAPATO-IONA 138 KV L                      | Rebuild line with higher-capacity conductor | \$1,872,000   | 100% (\$1,872,000)    |
| JARRETT SUBSTATION-UNION CARBIDE 138 KV L | Rebuild line with higher-capacity conductor | \$2,527,000   | 100% (\$2,527,000)    |
| CURECANTI-ELIZA CRK 345 KV                | Reconductor line                            | \$6,679,000   | 6% (\$424,000)        |
|                                           | Total allocated to project                  |               | \$82,621,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.